

PYTHON PROGRAMMING PROJECT

Super Secret Messages

Book Cipher



Gaël KERNINON & Clément GOMEZ

Cracow University of Technology

May 29, 2026

INTRODUCTION

The Rebecca Code

Our project was inspired by WWII espionage techniques, where secret agents used famous novels as encryption keys.

 **Goal:** Create a cryptographic tool in Python.

 **Key:** The book itself (*The Lost World*) and a integer (*Switch*)

 **Techniques:** Uses file processing, dictionaries, lists, and pseudo-random generation



ENCODING PROCESS



Preprocessing and Dictionaries

The algorithm firstly remove punctuation and then create two dictionaries, `D_words` and `D_letters`, for index storage.



Randomized Selection

For each word, the program randomly picks one index among all its occurrences. This makes frequency analysis impossible.



The Switch Mechanism

A secret numerical offset is added to the indices. Even with the correct book, an adversary will fail without this secondary key.

HANDLING MISSING WORDS :

THE LETTER MODE

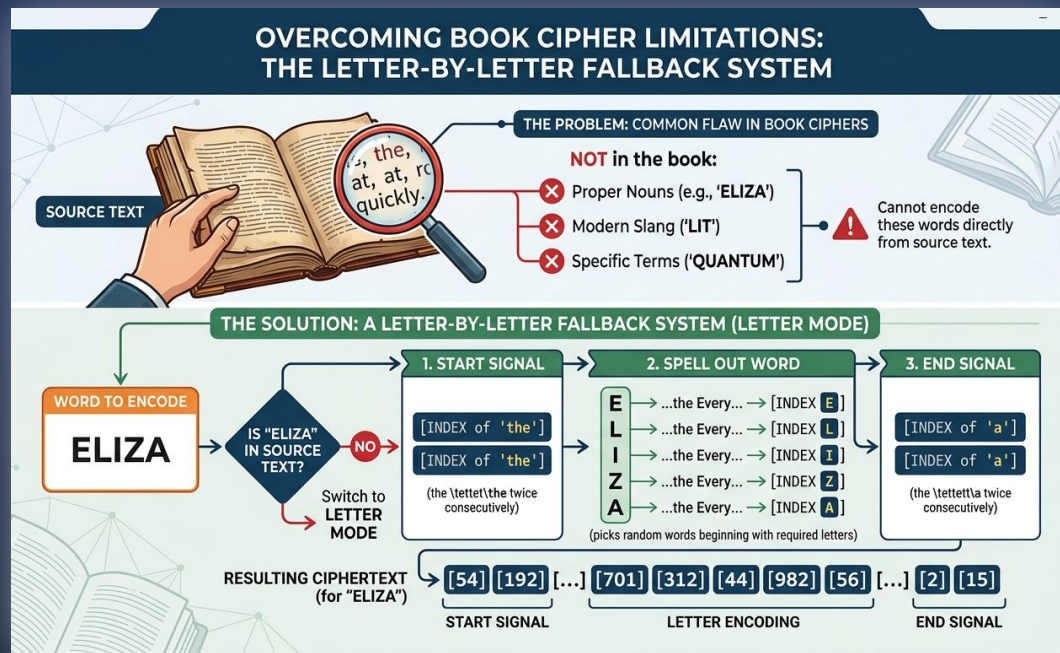
How do we encrypt words missing from the book ?

(Proper Nouns, Modern Slang) ?

Markers: The sequence "the the" initiates spelling mode and the sequence "a a" terminates it.

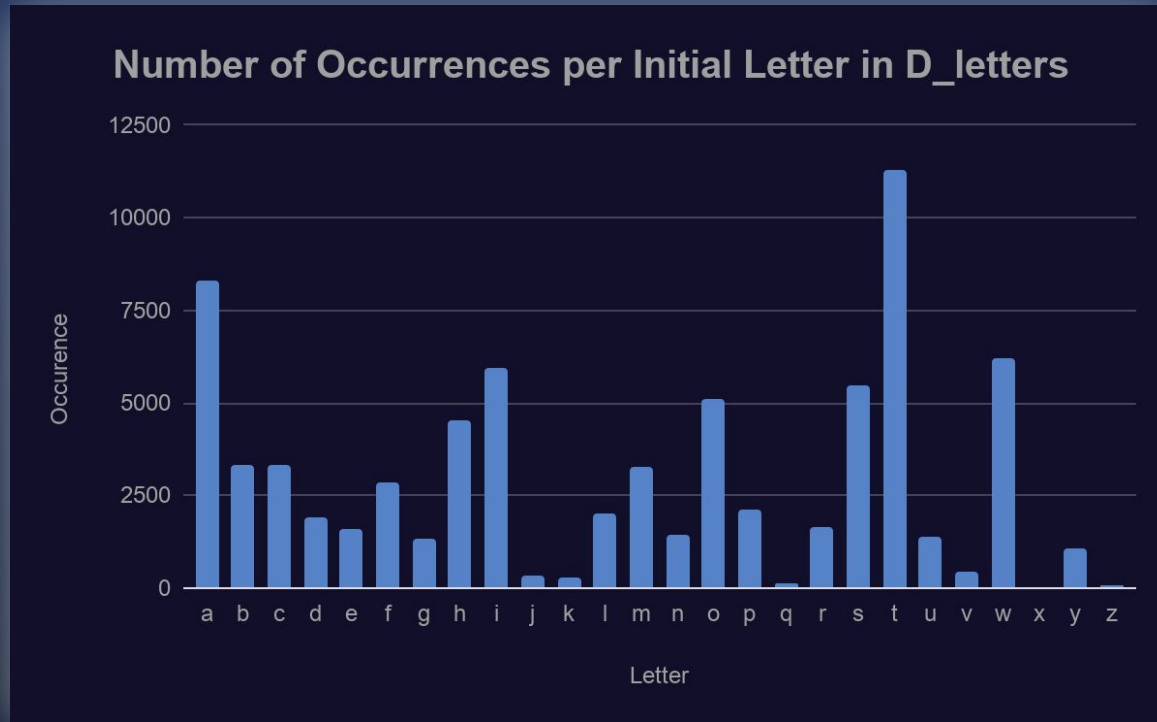
Spelling: The word is spelled out using the first letters of random words from the text.

Example: "ELIZA" becomes a sequence of indices for words starting with E, L, I, Z, A.



Flowchart and Example of Encoding Missing Word ELIZA
(AI-Generated Image)

IMPLEMENTATION LIMITS



Analysis of Letter Frequencies in The Lost World source book

DECODING PROCESS



Remove the Switch

Each encoded integer is shifted back by subtracting the predefined switch value used during the encryption process.



Detect Letter Mode

The decoder automatically detects special “Letter Mode” sequences used to encode unknown or missing words letter by letter.



Reconstruct the Message

Decoded indices are converted back into words in order to progressively rebuild the original plaintext message.

